

Surface Mesh Analysis Pipeline for Al(Ga) Grain Face Characterization

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1 Introduction

This report describes a Modula-3 library suite for analyzing confocal microscope surface data of aluminum–gallium alloy samples. The data originates from an Olympus LEXT laser scanning confocal microscope, which produces 3D surface height maps stored as binary PLY mesh files. Individual grain faces were previously segmented by hand in CloudCompare v2.14; the pipeline described here automates the segmentation and adds quantitative curvature and anomaly analysis.

The pipeline addresses three questions:

1. What are the shapes of individual crystal grain faces (orientation, curvature, deviation from planarity)?
2. Where are particle inclusions and surface defects located?
3. How does the surface roughness decompose across spatial frequency bands?

All code is written in Modula-3, built with the CM3 compiler, and lives under `Al_Ga/` in the `m3utils` tree. Visualization uses `gnuplot`.

2 Data

2.1 Instrument

The data was acquired on an Olympus LEXT OLS-series laser scanning confocal microscope at $550\times$ magnification. The instrument produces a 3D surface height map by scanning a laser spot across the sample and using confocal optics to measure the height at each pixel. Native coordinates are in nanometers.

2.2 File format

Data is stored in the PLY (Polygon File Format) binary little-endian format. Two types of files are present:

Full surface scans. 1024×1024 vertex grids ($\sim 1\text{M}$ vertices, $\sim 2\text{M}$ triangular faces) with three float properties per vertex (x, y, z). These are regular height maps downsampled from the raw LEXT output. Six samples exist, one per measurement site.

Pre-segmented face meshes. Irregular triangle meshes (50K–324K vertices) extracted from the surface scans using CloudCompare v2.14. These have nine float properties per vertex: position (x, y, z), surface normal (n_x, n_y, n_z), and original coordinates. Some files also include per-vertex RGB color as unsigned-byte properties. A total of 38 face meshes exist across four sub-samples (550_1.1 through 550_1.4), with 8–12 faces per sub-sample.

2.3 Samples

The dataset comprises sub-samples labeled 550_1.1 through 550_1.4 (detailed face meshes with grain-face measurements) and six surface scans labeled 550_1 through 550_6 in the `Data.AlGa.samples_-Bevis` directory.

3 Pipeline Architecture

The pipeline consists of five libraries and two programs. Each library is a separate CM3 package under `Al_Ga/`.

3.1 Package overview

Package	Type	Description
<code>plylib</code>	library	Binary PLY reader. Handles <code>float</code> and <code>uchar</code> vertex properties, triangular faces. Float properties are stored in a flat array; <code>uchar</code> properties (e.g. RGB) are skipped.
<code>meshlib</code>	library	3D vector operations (<code>Vec3</code>) and triangle mesh primitives (<code>TriMesh</code>): per-face normals and areas, area-weighted vertex normals, compressed 1-ring adjacency, centroid, mean normal, total area.
<code>facetlib</code>	library	Plane fitting (area-weighted mean normal), Rodrigues rotation to z -up frame, height field extraction. Region-growing segmentation with priority-queue seed selection and tilt-angle filtering.
<code>curvlib</code>	library	Cotangent-Laplacian mean curvature (units $1/\text{length}$), quadratic surface fitting, residual-based anomaly detection. Height field gridding with aspect-ratio-matched cells, Gaussian multiresolution smoothing, band-pass decomposition.
<code>plydemo</code>	program	End-to-end command-line tool: single-face analysis, full-surface segmentation, multiresolution contour output.
<code>plytest</code>	program	Diagnostic tool for PLY file inspection.

3.2 Build system

All packages use the standard CM3 build conventions: each has `src/m3makefile` and `src/m3overrides`. A shared `Al_Ga/m3overrides` sets up cross-package references and enables static linking (`build_standalone()`). The `facetlib` package instantiates a generic priority queue from `libm3`'s `PQueue` generic with an integer priority type for seed selection during segmentation.

3.3 Dependencies

The pipeline depends only on the CM3 standard libraries (`libm3`) and the existing `matrix` package in `m3utils` (for potential future use—currently the pipeline uses its own 6×6 Gaussian elimination for the quadratic fit, but the `matrix` package's SVD is available for more sophisticated fitting). The `Vec3` module provides lightweight 3D vector operations in `LONGREAL`.

4 Algorithms

4.1 Plane fitting and coordinate rotation

Given a triangle mesh representing a grain face, the best-fit plane is determined by the area-weighted mean normal:

$$\hat{\mathbf{n}} = \frac{\sum_f A_f \hat{\mathbf{n}}_f}{\left\| \sum_f A_f \hat{\mathbf{n}}_f \right\|}$$

where A_f and $\hat{\mathbf{n}}_f$ are the area and unit normal of face f . The centroid is likewise area-weighted.

The rotation matrix R that maps $\hat{\mathbf{n}}$ to $(0, 0, 1)$ is computed via Rodrigues' formula. Let $\mathbf{k} = \hat{\mathbf{n}} \times \hat{\mathbf{z}}$ (normalized), and $\cos \theta = \hat{\mathbf{n}} \cdot \hat{\mathbf{z}}$. Then:

$$R = I + (\sin \theta) K + (1 - \cos \theta) K^2$$

where K is the skew-symmetric matrix of $\mathbf{k}$. Special cases for $\hat{\mathbf{n}} \approx \pm \hat{\mathbf{z}}$ are handled separately.

After rotation, vertex positions are expressed as (x, y, z) where z is the height above the best-fit plane.

4.2 Curvature estimation

Mean curvature is computed using the cotangent-weighted discrete Laplace–Beltrami operator. For a vertex i with 1-ring neighbors $j \in N(i)$, the discrete mean curvature is:

$$H_i = \frac{1}{2A_i} \sum_{j \in N(i)} (\cot \alpha_{ij} + \cot \beta_{ij}) (z_j - z_i)$$

where α_{ij} and β_{ij} are the angles opposite edge (i, j) in the two adjacent triangles, and A_i is the mixed Voronoi area (approximated here as one-third of the total incident face area). This formula has units of 1/length—the inverse radius of curvature.

The cotangent of an angle between vectors $\mathbf{a}$ and $\mathbf{b}$ is computed as:

$$\cot \theta = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a} \times \mathbf{b}\|}$$

For display, curvature values are converted from 1/nm (the native mesh unit) to mm^{-1} by multiplying by 10^6 .

4.3 Quadratic surface fitting and residuals

The height field $z(x, y)$ is fit to a quadratic model:

$$\hat{z}(x, y) = ax^2 + bxy + cy^2 + dx + ey + f$$

The six coefficients are determined by least squares, solving the 6×6 normal equations $A^T A \mathbf{c} = A^T \mathbf{b}$ via Gaussian elimination with partial pivoting. The residual at each vertex is $r_i = z_i - \hat{z}(x_i, y_i)$.

4.4 Anomaly detection

Anomalous vertices (candidate particle inclusions or surface pits) are identified as those whose residual exceeds a threshold number of standard deviations:

$$|r_i - \bar{r}| > k \sigma_r$$

where $\bar{r}$ and σ_r are the mean and standard deviation of the residuals, and k is a user-specified threshold (default $k = 3$).

4.5 Region-growing segmentation

Full surface scans are segmented into planar grain faces by region-growing. The algorithm:

1. Compute area-weighted vertex normals for the entire mesh.
2. Initialize a priority queue containing all vertices, with priority equal to the negative number of neighbors (so interior vertices with many neighbors are extracted first).
3. Extract the highest-priority unclaimed vertex as a seed.
4. BFS-grow the region: add unclaimed neighbors whose vertex normal is within an angular threshold $\theta_{\max}$ of the running mean normal of the region.
5. When growth stops, record the region. Repeat from step 3 until the priority queue is exhausted.
6. Discard regions with fewer than a minimum number of vertices.
7. Discard regions whose mean normal deviates more than a specified angle from the vertical (z -axis), retaining only top-facing grain faces.
8. Sort remaining regions by decreasing vertex count.

Default parameters: $\theta_{\max} = 10^\circ$, minimum vertices = 500, maximum tilt from vertical = 30° .

4.6 Multiresolution height field decomposition

The height field is decomposed into spatial frequency bands by Gaussian smoothing at geometrically spaced scales.

4.6.1 Gridding

Scattered vertex (x, y, z) data is binned onto a regular grid with cell dimensions chosen to match the aspect ratio of the data bounding box while approximating a target total cell count. For a bounding box of width W and height H with target N cells:

$$\Delta = \sqrt{WH/N}, \quad n_x = \lceil W/\Delta \rceil, \quad n_y = \lceil H/\Delta \rceil$$

Cells with multiple vertices are averaged. Cells outside the face boundary (no vertex data) are masked; for smoothing purposes, empty cells are filled by iterative nearest-neighbor interpolation, but masked cells are written as NaN in the output so that gnuplot omits them from the display.

4.6.2 Gaussian smoothing

A separable Gaussian filter with standard deviation σ (in grid cells) is applied in two passes (horizontal, then vertical). The kernel is truncated at 3σ and normalized. Boundary cells are clamped (extended by replication).

4.6.3 Band-pass decomposition

Given L smoothing levels with geometrically spaced σ values from $\sigma_{\max}$ (coarsest) to $\sigma = 0$ (raw data), the band-pass at level k is:

$$B_k(x, y) = S_{k+1}(x, y) - S_k(x, y)$$

where S_k is the smoothed field at scale σ_k . The coarsest level S_0 captures the overall bowl/saddle shape; each band B_k captures progressively finer spatial detail. This is a discrete approximation to a Laplacian pyramid.

5 Results

5.1 Automatic segmentation

The full surface scan `AlGa_LEXT_surface_downsample_1024_6.ply` (1,048,576 vertices, 2,093,058 faces) was segmented with parameters $\theta_{\max} = 10^\circ$, minimum 500 vertices, maximum tilt 30° . Nine top-facing grain regions were identified:

Region	Vertices	Tilt ($^\circ$)	Normal
0	4787	4.4	(−0.073, −0.020, 0.997)
1	2440	5.9	(0.010, −0.103, 0.995)
2	1535	24	(0.069, 0.406, 0.911)
3	1373	1.0	(−0.009, 0.015, 1.000)
4	1274	1.2	(0.003, 0.021, 1.000)
5	765	27	(0.038, −0.448, 0.893)
6	748	3.8	(0.005, −0.066, 0.998)
7	651	6.3	(0.004, −0.110, 0.994)
8	506	7.0	(−0.117, 0.035, 0.993)

Regions 3 and 4, with tilt angles of 1.0° and 1.2° respectively, are nearly perfectly horizontal grain faces. Regions 2 and 5, at 24° and 27° , are tilted but still top-facing. Steep sidewall faces (tilt $> 30^\circ$) were automatically excluded.

5.2 Per-region curvature and anomalies

Each extracted region was individually analyzed. The table below shows the height span (peak-to-peak after plane subtraction), residual standard deviation (after quadratic fit), and number of detected anomalies at 3σ threshold:

Region	Height span (μm)	Residual σ (μm)	Anomalies
0	676	58.5	52
1	433	38.2	38
2	423	48.2	13
3	1050	182	2
4	1360	89.1	22
5	152	14.4	9
6	701	41.9	11
7	214	10.9	16
8	128	16.5	3

Region 5, with the smallest height span ($152\text{ }\mu\text{m}$) and residual σ ($14.4\text{ }\mu\text{m}$), is the flattest face. Region 3 has the largest residual ($182\text{ }\mu\text{m}$) but only 2 detected anomalies, suggesting a smoothly curved rather than rough surface.

5.3 Curvature map with anomaly overlay

Figure 1 shows the cotangent-Laplacian mean curvature map for Face 5 of sample 550_1_1 (128,549 vertices), with detected anomalies overlaid as black circles. Curvature is displayed in mm^{-1} ; the color scale is clipped at $\pm 2\text{ mm}^{-1}$ (corresponding to a radius of curvature of $500\text{ }\mu\text{m}$).

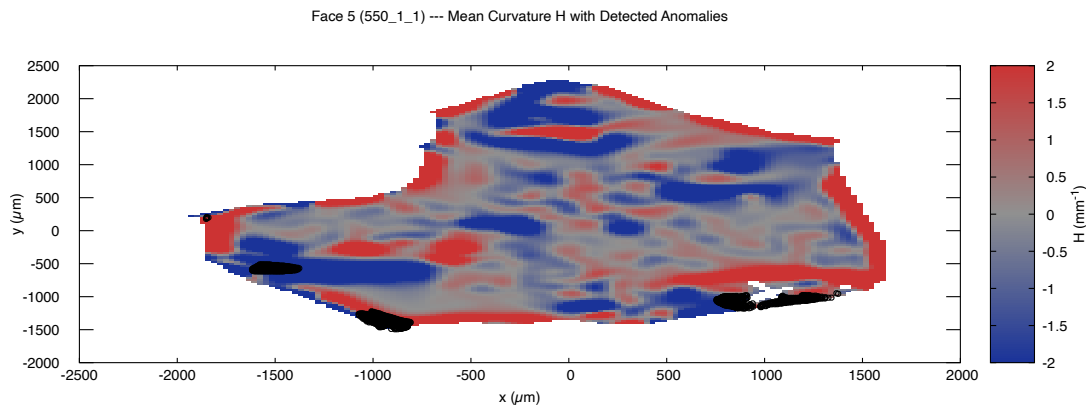


Figure 1: Mean curvature H (mm^{-1}) for Face 5, with anomalies (3σ residual threshold) marked as black circles. Gray indicates near-zero curvature; red is convex, blue is concave. Color palette follows the CIE Luv perceptual uniformity recommendations of Nystrom (2008).

5.4 Spectral decomposition

Figure 2 shows the multiresolution decomposition of the same face into six frequency bands. The coarsest level captures the overall shape (a gentle bowl with $\sim 500\text{ }\mu\text{m}$ peak-to-peak). Successive bands reveal medium-scale undulations and fine-scale roughness. The finest band (Band 4) shows features at the grid resolution ($\sim 5\text{ }\mu\text{m}$), where particle inclusions and polishing artifacts are most visible.

5.5 Height field

Figure 3 shows the raw height field for Face 5 after rotation to the z -up frame. The irregular face boundary is preserved through masking; cells outside the face are omitted.

6 Usage

6.1 Building

```
cd Al_Ga/plylib/src && cm3 -override
cd Al_Ga/meshlib/src && cm3 -override
cd Al_Ga/facetlib/src && cm3 -override
```

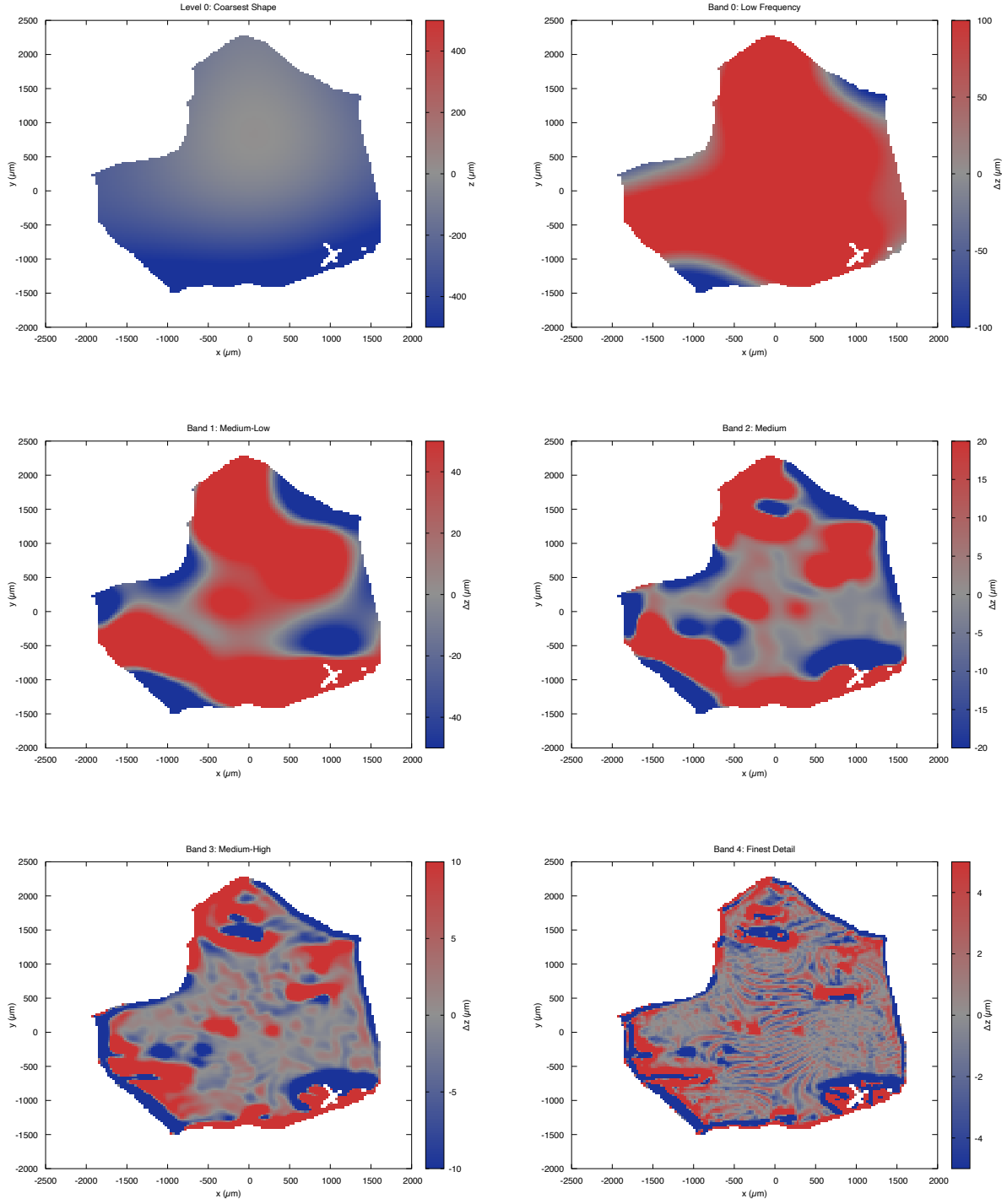



Figure 2: Multiresolution height field decomposition of Face 5. Top-left: coarsest smoothed level (Gaussian $\sigma = 8$ cells). Remaining panels: band-pass differences between successive smoothing levels, from low frequency (Band 0) to the finest detail (Band 4). Color scales decrease by approximately $2\times$ per band, consistent with a power-law roughness spectrum. All values in μm .

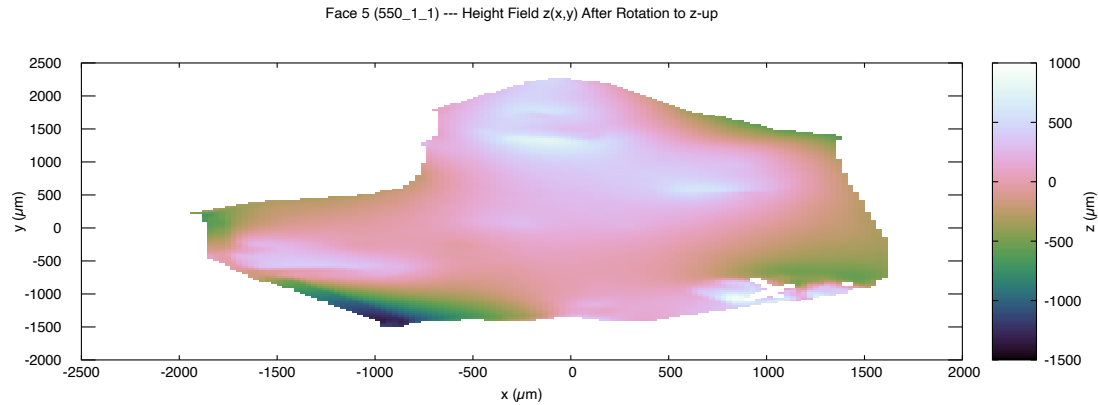


Figure 3: Height field $z(x, y)$ for Face 5 after best-fit plane subtraction and rotation. Units: μm . The irregular grain face boundary is visible.

```
cd Al_Ga/curvlib/src && cm3 -override
cd Al_Ga/plydemo/src && cm3 -override
```

6.2 Single face analysis

```
plydemo 550_1_1/Face_1_mesh.ply
```

Prints mesh statistics, curvature summary, and detected anomalies to standard output.

6.3 Segmentation of a full surface scan

```
plydemo -segment -angle 10 -minverts 500 -maxtilt 30 \
AlGa_LEXT_surface_downsample_1024_6.ply
```

Segments the surface into planar grain faces, filters by tilt angle, and reports per-region curvature and anomaly statistics.

6.4 Multiresolution contour output

```
mkdir outdir
plydemo -contour outdir -levels 6 550_1_1/Face_5_mesh.ply
gnuplot outdir/face_contours.gp
```

Writes gnuplot-compatible grid data files for height levels, band-pass differences, curvature, residuals, and anomaly points. All coordinates are output in micrometers; curvature in mm^{-1} .

7 Color

The color palette for curvature maps follows the recommendations of the CIE Luv perceptual uniformity analysis in the “Curves Reference Manual” (Nystrom, 2008): a diverging palette with a neutral gray midpoint (#909090) and blue/red endpoints chosen to be approximately equidistant in Luv space. This avoids the banding artifacts of HSV-based palettes and ensures that equal curvature differences produce equal perceptual differences.

8 Future Work

- **FFT-based spectral decomposition.** Replace the Gaussian pyramid with a proper 2D FFT, enabling quantitative power spectral density analysis and identification of preferred roughness wavelengths.
- **Anomaly clustering.** Group individual anomalous vertices into connected clusters, compute per-cluster area, depth, and centroid for statistical characterization of inclusion size distributions.
- **PLY output.** Write segmented regions and curvature maps as PLY files with scalar vertex properties, for visualization in CloudCompare.
- **Gaussian curvature.** Add the angle-deficit estimator for Gaussian curvature K , enabling classification of surface shapes (elliptic, hyperbolic, parabolic).
- **Batch processing.** Process all six surface scans and produce a PDF atlas of all grain faces with curvature maps and spectral decompositions.
- **Voronoi area.** Replace the $A_i/3$ barycentric approximation with the proper mixed Voronoi area of Meyer et al. (2003) for more accurate curvature at obtuse triangles.

References

- Meyer, M., Desbrun, M., Schröder, P., and Barr, A.H. (2003). “Discrete differential-geometry operators for triangulated 2-manifolds.” *Visualization and Mathematics III*, Springer, pp. 35–57.
- Nystrom, M. (2008). “Curves Reference Manual.” Generation Capital Ltd. Chapter 4: Color (CIE Luv perceptual uniformity for quantitative data visualization).